

Computer Graphics — Exam Questions and Answers

Part 1 : Fundamentals of Computer Graphics

Q1. Define Computer Graphics and list any four application areas.

Computer Graphics is the field of creating, editing, and displaying images using a computer.

Four application areas:

- Education and training
- Gaming and entertainment
- Medical imaging (scans, diagnosis)
- CAD / engineering design

Q2. Differentiate Interactive and Passive graphics.

Interactive graphics: the user can change the image in real time. Example: video games, CAD software.

Passive graphics: the user can only view the image, not change it. Example: a printed chart, a static slide.

Q3. List and explain the four main parts of a Display Processor.

- Display controller — manages data flow between CPU and display device.
- Frame buffer (refresh buffer) — memory that stores the pixel image.
- Digital-to-analog converter — turns stored pixel data into a signal the screen can show.
- CRT (or display screen) — shows the final image to the user.

Q4. Describe the components and working of a Raster Scan System.

Components: CPU, frame buffer, video controller, monitor, keyboard/input devices.

Working: the image is stored as pixels in the frame buffer. The video controller reads this data and refreshes the screen row by row, moving the electron beam left to right, top to bottom, many times per second.

Q5. Compare Random Scan and Raster Scan displays.

Heading	Random Scan	Raster Scan
Method	Beam draws only the lines of the picture (vector)	Beam scans the whole screen row by row
Refresh rate	Depends on number of lines (not fixed)	Fixed, usually 60 Hz
Resolution	High, smooth lines	Depends on pixel count, can look jagged
Advantages	Smooth, clean lines, no jagged edges	Can show realistic, filled, shaded images
Disadvantages	Cannot fill solid areas easily	Lines can look jagged (staircase effect)
Example	Old vector graphics terminals	Modern TVs and computer monitors

Q6. Explain Beam Penetration and Shadow-Mask methods.

Beam Penetration: the screen has two phosphor layers, red inside and green outside. A slow beam excites only the red layer, a fast beam reaches the green layer, and speeds in between give mixed colors. It gives only a few colors and is used in random-scan monitors.

Shadow-Mask: uses three electron guns (red, green, blue) and a metal mask with tiny holes. The mask lines up each beam with the correct color phosphor dot. This gives a wide range of colors and is used in raster-scan color monitors.

Q7. Short notes: (a) DVST (b) Plasma Panel (c) LCD.

(a) DVST (Direct View Storage Tube): stores the picture as a charge on a wire mesh behind the screen, so it does not need refreshing. Downside: cannot erase just one part, and it shows no color.

(b) Plasma Panel Display: uses neon gas between two glass plates. Voltage makes the gas glow to form the picture. It is a flat panel and does not need refreshing, but it is costly.

(c) LCD (Liquid Crystal Display): uses liquid crystals that change how light passes through when voltage is applied. It is flat, thin, and uses low power. Common in laptops and monitors.

Q8. List the five main components of a CRT.

- Electron gun
- Focusing system
- Deflection system (yoke)
- Phosphor-coated screen
- Electron beam

Q9. What is Scan Conversion? Explain the DDA line algorithm.

Scan Conversion is the process of converting a continuous graphics shape (like a line) into discrete pixels so it can be shown on a raster display.

DDA (Digital Differential Analyzer) algorithm: it calculates points along a line using small equal steps in x and y (based on the slope), using floating-point addition at each step.

Advantage: simple and easy to code.

Disadvantage: uses floating-point math, which is slower and can build up rounding errors.

Q10. Differentiate DDA and Bresenham's line algorithm. Why is Bresenham preferred?

DDA uses floating-point calculations and rounding at every step. Bresenham's algorithm uses only integer addition and comparison.

Bresenham is preferred because it is faster and more accurate, with no rounding error build-up.

Q11. Explain eight-way symmetry of a circle in Bresenham's Circle algorithm.

A circle is symmetric in 8 equal parts (octants). So the algorithm only calculates points for one octant (45°). For each point (x, y) found, the other 7 points are obtained directly by swapping and flipping signs: $(\pm x, \pm y)$ and $(\pm y, \pm x)$. This cuts the calculation work by 8 times.

Q12. What is the Midpoint Ellipse Algorithm? Describe its two regions.

It is a method to draw an ellipse using only integer math, by checking a decision parameter at the midpoint between two candidate pixels.

Region 1: where the slope of the ellipse boundary is greater than -1. Here x is incremented by 1 each step.

Region 2: where the slope is less than -1. Here y is incremented by 1 each step.

Q13. Differentiate Boundary Fill and Flood Fill algorithms.

Boundary Fill: fills an area by spreading out from a start point until it hits a specified boundary color. Works best when the whole boundary is one single color.

Flood Fill: fills an area by replacing one specific interior color with a new fill color. Useful when the region is bounded by more than one color.

Q14. List the five basic 2D transformations. Explain Rotation about a pivot point.

- Translation
- Rotation
- Scaling

- Shearing
- Reflection

Rotation about a pivot point (not the origin): first translate the pivot point to the origin, then rotate the object by the needed angle, then translate it back to the original pivot position.

Q15. Differentiate Geometric transformation and Coordinate transformation.

Geometric transformation: the object itself moves, resizes, or rotates, while the coordinate system stays fixed.

Coordinate transformation: the object stays fixed, but the coordinate system (axes) is moved or rotated.

Q16. Define Window, Viewport, Zooming, and Panning.

Window: a rectangular area in world (scene) coordinates that is chosen for viewing.

Viewport: the area on the actual screen where the window's contents are displayed.

Zooming: making the visible area larger or smaller to enlarge or shrink the view.

Panning: moving the window across the scene to see different parts, without changing zoom level.

Q17. What is Clipping? Give four applications.

Clipping is the process of cutting off the parts of a picture that fall outside a chosen window, so only the visible part is drawn.

- Viewing transformations (showing only the window area)
- Identifying visible surfaces in 3D scenes
- Drawing shapes such as windows and menus
- Creating objects using set operations (like cutting shapes)

Q18. Explain the Elastic/Rubber Band and Dragging techniques.

Elastic (Rubber Band) technique: one end of a shape (like a line) is fixed, and the other end follows the mouse pointer as it moves, like stretching a rubber band, letting the user see the shape before finalizing it.

Dragging technique: the user selects an object with the pointer and moves it across the screen interactively, while the object keeps its shape.

Part 2 : Shading

Q19. Define Shading. Explain Ambient, Diffuse, and Specular illumination.

Shading is the process of giving color and light intensity to a surface based on light, so a 3D object looks solid and realistic.

Ambient light: constant background light present everywhere in the scene, with no direction, giving base brightness.

Diffuse light: light that scatters equally in all directions off a rough/matte surface; brightness depends on the angle between the light and the surface normal.

Specular light: the shiny highlight on a surface; it depends on the angle between the viewer and the reflected light direction.

Q20. Compare Flat, Gouraud, and Phong shading.

Heading	Flat	Gouraud	Phong
Basis	One color per polygon face	Color at vertices, interpolated	Normals interpolated, lit per pixel
Calculation stage	Once per polygon	At each vertex	At each pixel
Realism	Low, flat-looking	Medium	High, smooth highlights
Speed	Fastest	Medium	Slowest

Q21. Key difference between Gouraud and Phong shading.

Gouraud shading calculates light only at the vertices and blends (interpolates) the colors across the polygon, so it can miss specular highlights in the middle of a face.

Phong shading interpolates the surface normal itself and calculates lighting at every pixel, giving more accurate and smoother highlights, at the cost of more computation.

Q22. Short notes on any four: Texture Mapping, Bump Mapping, Environment Mapping, Shadow Mapping, Ray Tracing, Global Illumination, Subsurface Scattering, PBR, BRDF.

Texture Mapping: wraps a 2D image onto a 3D surface to add visual detail without adding extra geometry.

Bump Mapping: fakes small surface bumps or wrinkles by changing surface normals, without changing the real geometry.

Shadow Mapping: renders the scene from the light's point of view to build a depth map, then uses it to work out which areas are in shadow.

Ray Tracing: traces light rays from the camera through each pixel into the scene to realistically simulate reflection, refraction, and shadows.

Q23. List the surface and illumination properties affecting shading.

Surface properties: color, texture, roughness, reflectivity, transparency.

Illumination properties: light intensity, light direction/position, light type (ambient, diffuse, specular), and light color.

Part 3 : 3D in Computer Graphics

Q24. Who is the father of interactive computer graphics? His 1963 contribution.

Ivan Sutherland is considered the father of interactive computer graphics. In 1963 he created Sketchpad, the first program that let a user draw and directly interact with graphics on screen using a light pen.

Q25. How do 3D transformations differ from 2D, especially rotation?

2D objects lie flat on one plane, so rotation needs only one angle, around a single axis (z). 3D objects can turn in space, so rotation needs three angles, one each for the x, y, and z axes.

Q26. Compare the five types of 3D projections.

Heading	Parallel	Orthographic	Oblique	Isometric	Perspective
Projection lines	Parallel to each other	Parallel, perpendicular to plane	Parallel, at an angle to plane	Parallel, equal angles on axes	Converge at one point
Distortion	No size distortion	No distortion, true shape/size	Depth distorted	Minor angle distortion, no size loss	Objects farther away look smaller
Key feature	Preserves relative size	Shows true front/top/side views	Shows depth on one view	Equal scale on all 3 axes	Realistic depth (like human eye)
Applications	Engineering drawing	Technical/mechanical drawing	Furniture/pictorial drawing	CAD, technical illustration	Games, movies, realistic renders

Q27. Why is Perspective projection considered the most realistic?

Perspective projection uses lines that converge to a single point (center of projection), just like how the human eye and cameras see the world. This makes far away objects look smaller, matching real-life depth perception.

Part 4 : Rendering

Q28. Define Rendering.

Rendering is the process of generating a 2D image from a 3D scene model by computing light, color, shading, and texture for each pixel.

Q29. Trace the evolution of rendering techniques (1960s to present).

- 1960s — Wireframe models, simple line drawings.
- 1970s — Flat and Gouraud shading, hidden surface removal.
- 1980s — Phong shading, ray tracing introduced.
- 1990s — Radiosity, global illumination, better texture mapping.
- 2000s — Real-time rendering using GPUs and programmable shaders.
- 2010s to present — Real-time ray tracing, physically based rendering (PBR), AI-assisted rendering.

Q30. Characteristics of good rendering (R-E-F-A-I).

- Realism — looks natural and believable
- Efficiency — renders in reasonable time
- Flexibility — works for different scenes/materials
- Accuracy — light and color are physically correct
- Interactivity — can update in real time when needed

Q31. Classify rendering algorithms into categories with examples.

- Rasterization — scan conversion, z-buffer algorithm
- Ray Tracing — ray casting, path tracing
- Shading Models — flat, Gouraud, Phong shading
- Global Illumination — radiosity, photon mapping
- Anti-Aliasing — supersampling, MSAA
- Camera Effects — depth of field, motion blur, bloom

Part 5 : Graphics APIs: DirectX and OpenGL

Q32. List the six main components of DirectX and their purpose.

- Direct3D — renders 3D graphics
- Direct2D — renders 2D graphics
- DirectSound — handles audio playback
- DirectInput — reads input from keyboard, mouse, joystick
- DirectPlay — handles networking and multiplayer
- DirectShow — handles media/video playback

Q33. Describe the five stages of the OpenGL Rendering Pipeline.

1. Vertex Processing — transforms vertex positions (model to screen space)
2. Primitive Assembly — groups vertices into points, lines, or triangles
3. Rasterization — converts primitives into fragments (potential pixels)
4. Fragment Processing — computes color/lighting for each fragment
5. Framebuffer Output — final pixel tests (depth, blending) and display

Q34. Compare DirectX and OpenGL.

Heading	DirectX	OpenGL
Developer	Microsoft	Khronos Group

Platform	Windows and Xbox only	Cross-platform (Windows, Mac, Linux, mobile)
Year	1995	1992
3D component	Direct3D	OpenGL core itself
Shading language	HLSL	GLSL

Q35. List five key features of OpenGL.

- Cross-platform (runs on many operating systems)
- Free and open standard
- Large library of 2D/3D drawing functions
- Supports hardware (GPU) acceleration
- Uses GLSL for writing shaders

Short / One-liner Questions

Q1. Define Pixel / Pel.

A Pixel (Picture Element) is the smallest unit of a digital image or screen, holding one color value.

Q2. What is a Lookup Table (LUT)?

A LUT is a table that maps stored pixel values to actual color or intensity output, used for fast color conversion.

Q3. Difference between interlaced and non-interlaced raster scanning.

Interlaced: the screen refreshes odd lines, then even lines, in alternate passes, saving bandwidth but can flicker.

Non-interlaced (progressive): every line is refreshed in one single pass, giving a smoother image.

Q4. Define BRDF and PBR in one sentence each.

BRDF (Bidirectional Reflectance Distribution Function): a function that describes how much light reflects off a surface in each direction.

PBR (Physically Based Rendering): a rendering approach that uses real-world material and light properties to produce realistic images.

Q5. Why does 3D rotation need three factors while 2D needs only one?

In 2D, everything lies flat on one plane, so rotation only happens around one axis (z). In 3D, an object can turn around the x, y, and z axes, so three separate angles are needed.

Q6. Memory trick for the OpenGL pipeline stages.

Stages in order: Vertex, Primitive assembly, Rasterization, Fragment, Framebuffer. A simple memory sentence: "Very Pretty Rivers Flow Forward" (V-P-R-F-F).

Q7. Two types of coordinate system orientations in 3D geometry.

Right-handed coordinate system and Left-handed coordinate system.

Q8. List the six display device types mentioned in the notes.

Based on standard classification: CRT, DVST, Plasma Panel Display, LCD, LED Display, and Electroluminescent (flat panel) Display.

Note: your source notes did not spell out this exact list in the text shared with me, so please check it against your own notes.

Exam Tip: focus most on the comparison tables (Random vs Raster Scan, Flat/Gouraud/Phong shading, the five projections, DirectX vs OpenGL) and the OpenGL pipeline stages — these are the highest-yield topics.